

# Allan Kangethe

U.S. Citizen | Eligible to Obtain Secret Clearance | Washington DC–Baltimore Area  
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## SUMMARY

Software Engineer and M.S. candidate (Georgia Tech OMSCS, Artificial Intelligence specialization) with experience building full-stack applications and machine learning systems — from CNNs for medical image classification to LLM-powered applications. Strong foundation in Python, deep learning frameworks, and the full software development lifecycle. Comfortable collaborating on teams or working independently in remote or on-site settings.

## TECHNICAL SKILLS

**Machine Learning:** Convolutional Neural Networks (CNN), Deep Learning, Model Training & Evaluation, Data Pipelines, Large Language Models (LLM), Local Model Deployment (Ollama)

**Languages:** Python, Java, C++, C#, JavaScript, SQL, R, Assembly

**Frameworks & Tools:** TensorFlow, Keras, Flask, Django, Node.js, Git, JIRA, Linux, Hadoop, Tableau, AWS (exposure)

**Databases:** SQL, SQLite, NoSQL

**Concepts:** Software Development Lifecycle (SDLC), Agile, Object-Oriented Design, Data Structures & Algorithms, Testing & Debugging, REST APIs, Model Deployment

## EXPERIENCE

### Software Engineer Intern — WezeshaNet Technologies Ltd

Jun 2024 – Dec 2024

Remote Development Team

Nairobi, Kenya

- Developed and optimized C++ and Python software modules for client applications.
- Collaborated in Agile sprints, using Git for version control and participating in peer code reviews.
- Debugged critical issues, designed software updates, and authored technical documentation for deployment procedures.

### Technical Specialist — Verizon

Oct 2025 – Present

Full-time, On-site

Maryland

- Diagnose and resolve hardware, software, and connectivity issues for customers, translating technical concepts into clear, non-technical explanations.
- Maintain high reliability and customer-satisfaction metrics in a fast-paced, performance-driven environment.

## PROJECTS

### Socratic Feedback AI Tutor

Jan 2026 – May 2026

Python, Flask, GPT-OSS, Ollama | [github.com/Amkangethe/final-project](https://github.com/Amkangethe/final-project)

- Built a full-stack web application that delivers adaptive, Socratic-style feedback on coding problems, guiding learners toward solutions rather than providing direct answers.
- Integrated the GPT-OSS large language model served locally via Ollama, eliminating API costs and keeping all learner data private and on-premise.
- Implemented session logging to capture and review learner interactions, and built a test suite to validate model-response handling.

### Breast Cancer Detection — AI Prediction Model

Jan 2024 – Mar 2024

Python, TensorFlow, Keras, CNN | [github.com/Nalan12/AI4ALL\\_Project](https://github.com/Nalan12/AI4ALL_Project)

- Engineered an end-to-end deep learning pipeline using a convolutional neural network to classify breast tissue images as benign or malignant.
- Achieved 98% classification accuracy through iterative model tuning, integrated testing, and debugging.

### Full-Stack Social Media Platform

Sept 2024 – Dec 2024

Python, Flask, SQLite, JavaScript, HTML/CSS | [github.com/Amkangethe/CMSC447](https://github.com/Amkangethe/CMSC447)

- Built a GUI-based social media platform with user-facing features, a Flask backend, and REST APIs serving a dynamic frontend.
- Designed and integrated a SQLite database for persistent storage; worked in an Agile team using Git and JIRA to coordinate tasks.

## EDUCATION

### Georgia Institute of Technology — M.S. in Computer Science (OMSCS)

Expected May 2028

Specialization in Artificial Intelligence

Atlanta, GA

### University of Maryland, Baltimore County (UMBC) — B.S. in Computer Science

May 2025

Relevant coursework: Algorithms, Data Structures, Software Engineering, Databases, AI

Baltimore, MD

Activities: National Society of Black Engineers

## CERTIFICATIONS

IBM Introduction to Software Engineering (2025) • IBM Introduction to Cloud Computing (2025) • AWS Certified Cloud Practitioner (In Progress)